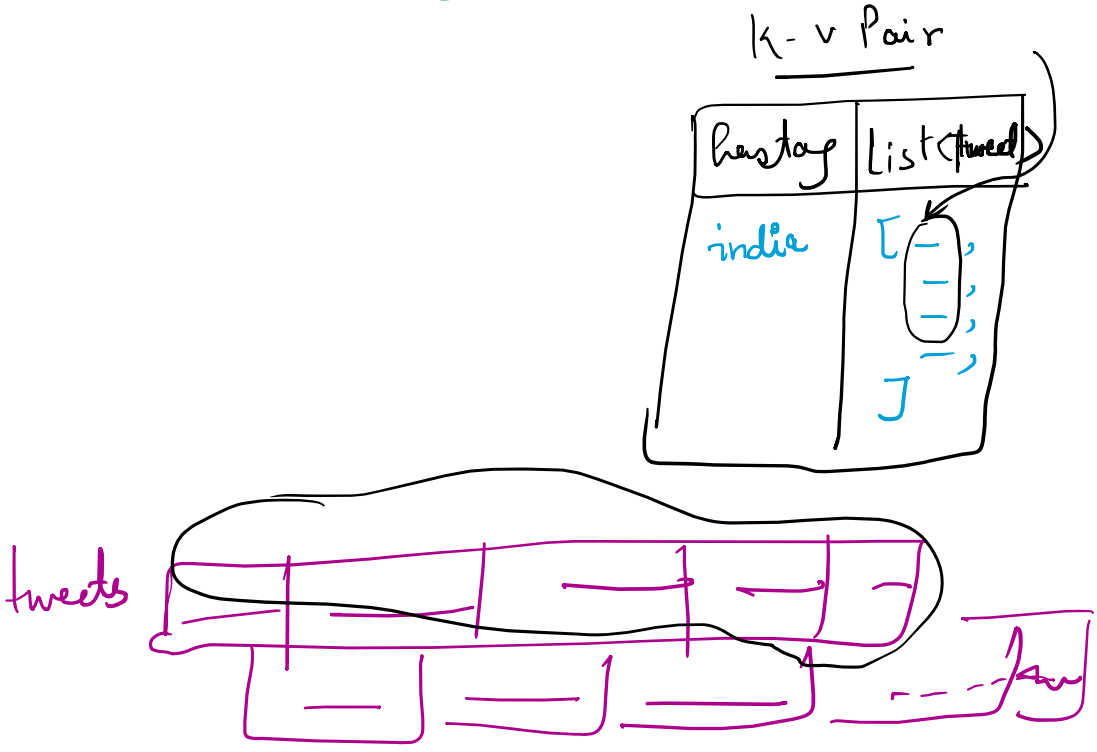
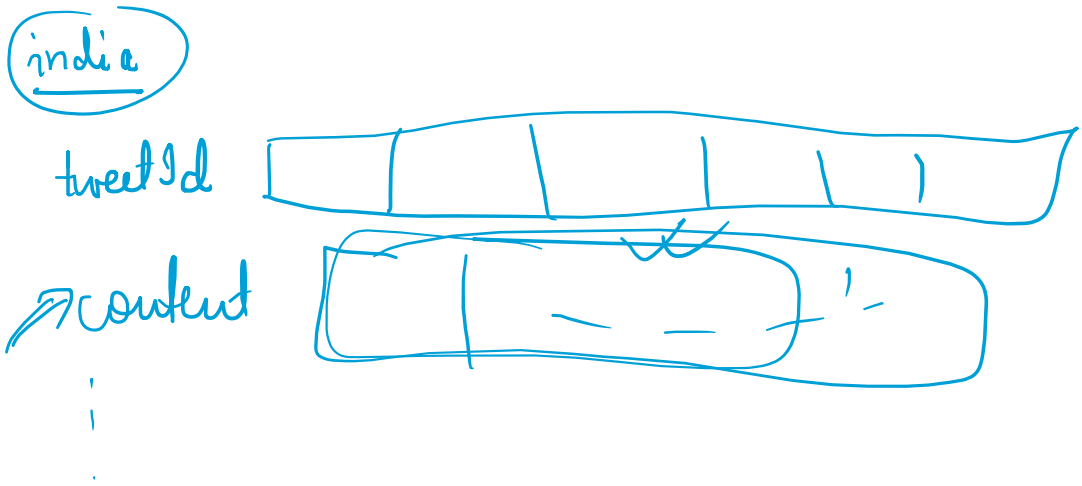
K-V	Document	Col^{n} - Family
→ When interested in value as a whole	→ When no fixed attribute	→ Query on col^{n}
→ Redis, DynamoDB	→ Mongo, Elastic search	→ Cassandra, HBase,

✱ Case

(1) Twitter Hashtag Storage

given a hashtag
fetch 10 latest tweets of that hashtag





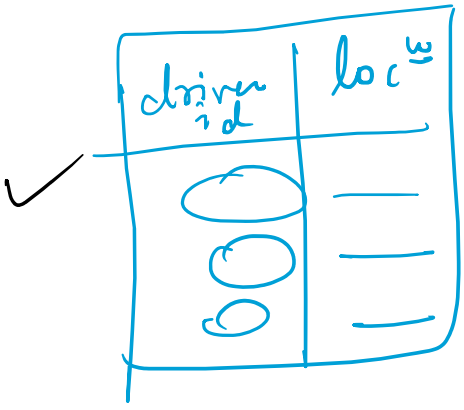
Q Get the score of a match !

Srh v/s mi ⟶ 420

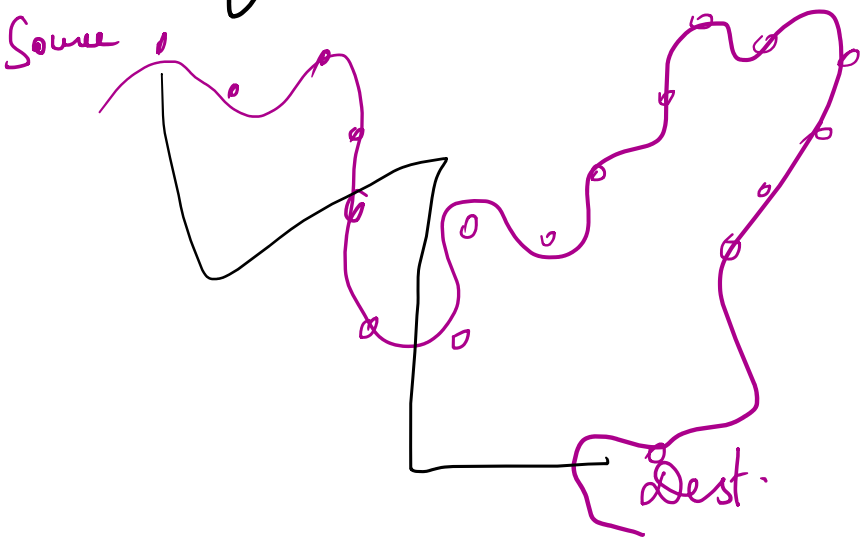




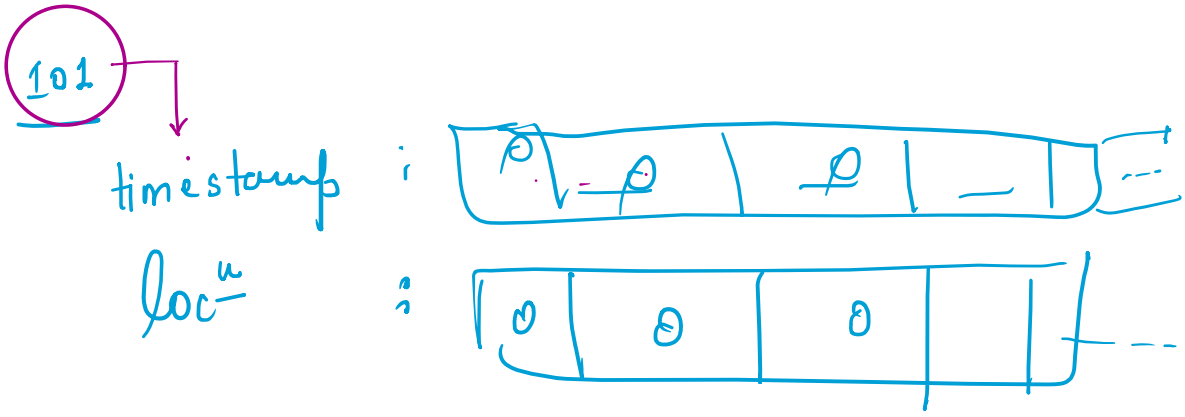
Q Current loc^{n} of a UBER driver !

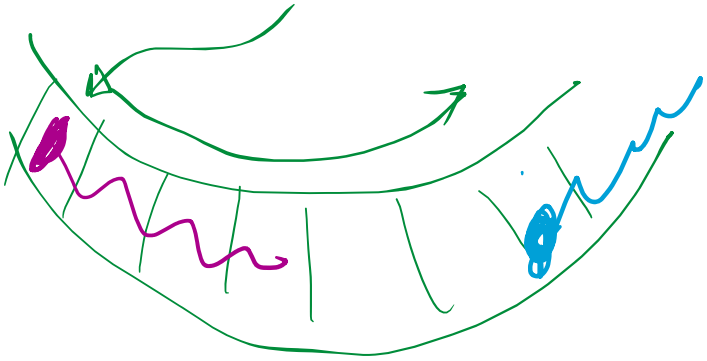


Loc^{n} history of a Uber Driver.



↳ Col^{n} family → for a driver_id





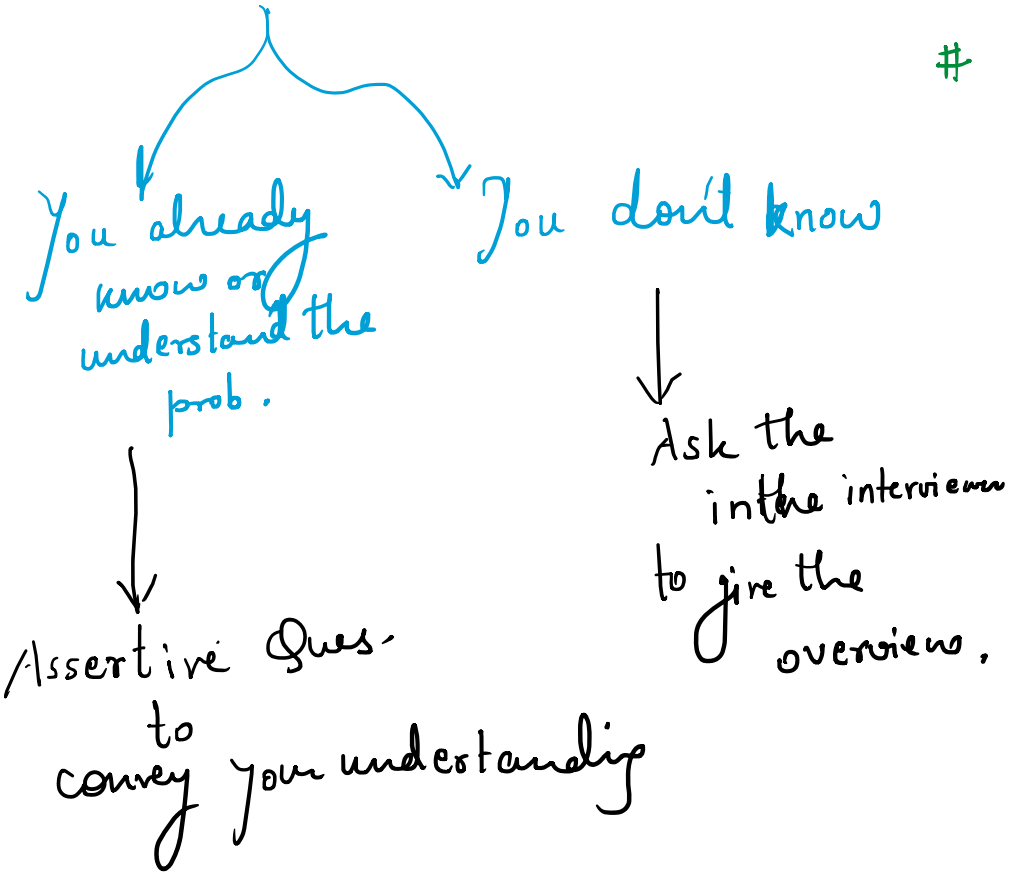
(*) Step By Step Approach.

HLD Problems

(*) Step 1 2y step Approach of HLD Problems.

0. Overview → Align w/ the interviewer.

#



1. Final-Req (Features that need to supported)

↳ don't waste too much time here.

↳ 3-5 mins.

Take a note of eventually agreed list.

↳ List down the MVP feature (Min^m viable product)

(Don't overthink / overdo) → Core Feature.

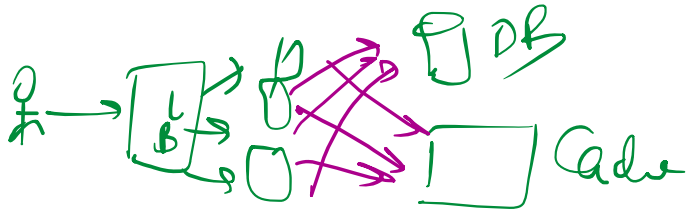
list: → (Don't overthink / overdo) → Core Feature.

2. Non-Fal Req.
   - ↳ expectations on how those features should work.
   - ↳ consistency v/s availability
   - ↳ consistency " latency.

3. Back of Envelope Estimation (BOE)
   - {Estimation of Scale}
   - ↳ Peak QPS
   - ↳ Size of data
   - ↳ Peak N/W BW
     - → Do you need sharding
     - → Read Heavy or Write Heavy

4. List APIs

5. Architecture Diag.



6. Deep Dive
   - (brainstorming on a particular component of the system)
   - How will I handle write queries
   - How will you handle recovery etc.

9:36 pm

* System Design — WhatsApp Messenger

* Overview → Messaging Appl^{n} for diff. media.
  - Text

* F^{al} Req.
  1. Send & Receive messages.
  2. Only text messages

1.)

2.) Only text messages

3.) 1:1 chatting

4.) Retention policy of 30 days.

5.) Messages are stored in backend ✗

or only on device.

6.) Get my Recent Conversations

7.) Cx can store chats on their cloud storage =



(X) Non-fᵘⁿᶜ Req.

↳ Consistency ✗ v/s Availability.

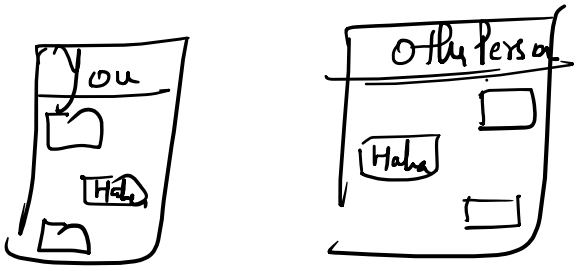
↳ Consistency ✓ v/s Latency

Consistency in context of WA Messenger

Consistency in context of WA Messenger

1. Send Message ()
   - → sender.
   - → Receiver

2. Order of Messages



⊛ Estimation of Scale.